



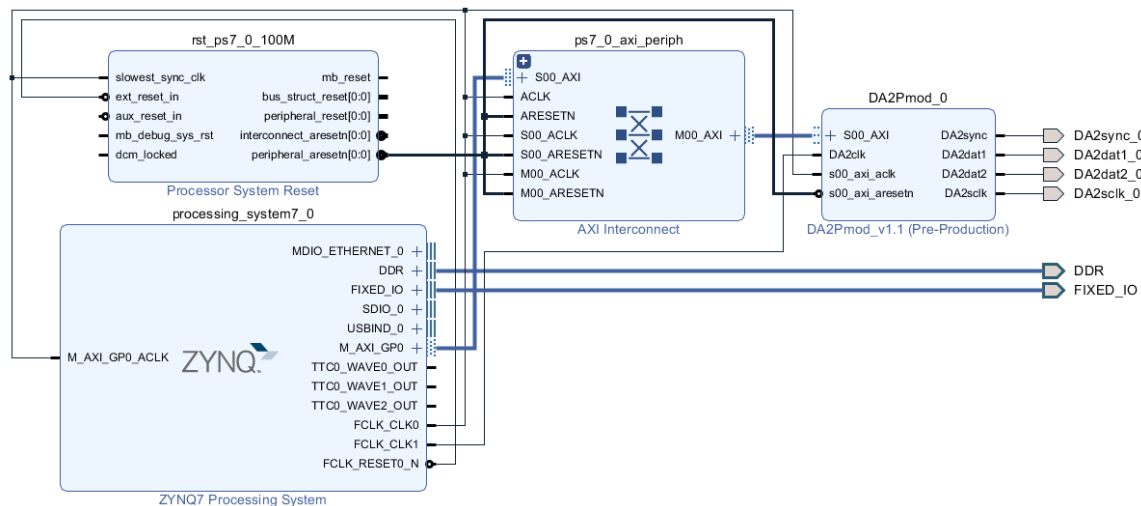
# ECE3623 Embedded System Design Laboratory



## PmodDA2 DAC Function Generator

Laboratory 6 is to configure the Digilent PmodDA2 digital-to-analog converter (DAC) and the Xilinx Vivado Zybo Board as a function generator. The DA2Pmod is a two channel, 12-bit DAC using the Texas Instruments (TI) DAC121S101 device. The Diligent reference manual for the PmodDA2 and the TI specification sheet for the DAC121S101 is posted on Canvas.

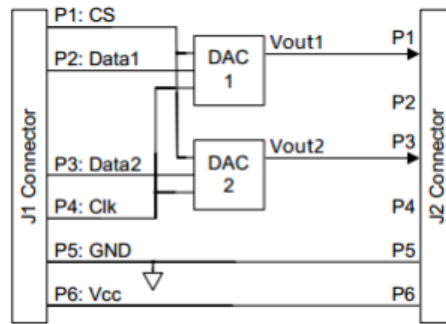
The Vivado hardware design uses the DA2Pmod IP block installed from the *ZynqPmodVivado-Library* as described in the *Zynq SPI Peripherals ADC and DAC* PowerPoints. The DA2Pmod SPI *sclk* (DA2sck) is connected to the Zynq processor clock FCK\_CLK1 at 50 MHz. The resulting Vivado block diagram is shown below.



The Vivado block design **constraint file DA2JE7\_10.xdc on Canvas** under *Pmod Vivado and SDK References* is required and provides the connection support for the DA2 DACPmod to pins 7 to 10 of the JE connector of the Zybo board.

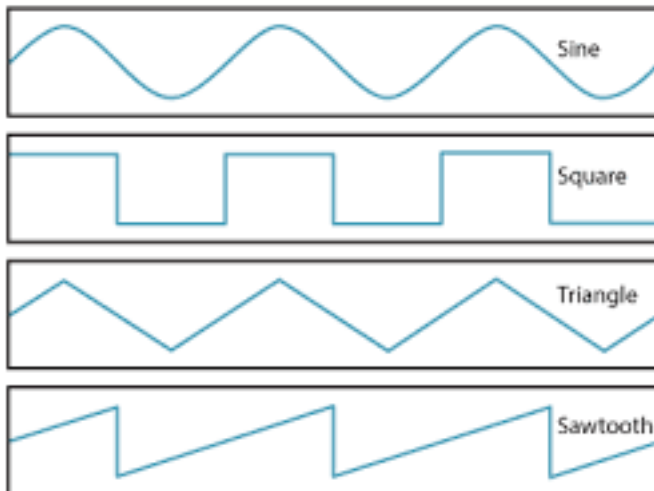
The PmodDA2 block diagram shows that pin 1 of J1 is the common active-low chip select ( $\sim$ CS) signal, pin 2 of J1 and pin 3 of J1 are the serial data for each of the DACs separately, and pin 4 of J1 is the serial clock (SCLK). Pin 5 of J1 is ground and pin 6 of J1 is Vcc.

Pin 1 of J2 is the DAC1 output signal. Pin 3 of J2 is the DAC2 output signal. Pin 5 of J2 is ground. Do not connect Pin 6 of J2 to anything since this is Vcc also.



Complete the Vivado hardware design above and launch SDK. Use the **DA2PmodTest.c** shown below.

DA2PmodTest.c generates a sawtooth waveform with data from 0 to 4095.

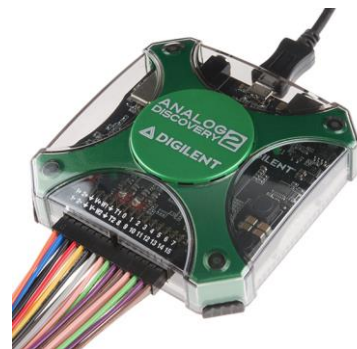


The output of the DA2Pmod DAC can be viewed on Channel 1 of the oscilloscope of the Digilent Discovery 2 using the Waveforms application. The reference for the Discovery 2 and Waveforms and Waveform 64-bit software are on Canvas.

The task for Laboratory 6 is as follows:

Modify the Vivado hardware and SDK software application for the DA2Pmod to output a triangular waveform from 0 to 4095.

Add a new GPIO to read the switches, which are used to set the amplitude and frequency of the triangular waveform follows:



SW0 = 0.5V amplitude at 4Hz;  
SW1 = 1V amplitude at 6Hz;  
SW2 = 1.5V amplitude at 8Hz;  
SW3 = 2V amplitude at 10Hz;

Note: the waveform rises/falls based on the speed of program execution. Therefore, you will likely have to use a sleep() delay between samples to achieve the desired frequencies.

Verify the DA2Pmod function generator output by measuring the four triangular waveforms using the Discovery 2 instrumentation.

```
// DA2PmodTest.c    ECE3622

#include "xparameters.h"
#include "xil_io.h"
#include "xil_printf.h"

//DAC2Pmod from Address Editor in Vivado
#define DA2acq      0x43C00000    //DA2 acquisition    - output
#define DA2dav      0x43C00004    //DA2 data available - input
#define DA2dat1     0x43C00008    //DA2 channel 1 data - output
#define DA2dat2     0x43C0000C    //DA2 channel 2 data - output

int main(void)
{
    int dacdata=0;    //DAC sawtooth data
    int dacdav;       //DAC data available
    int dacacq=0;     //DAC acquire

    xil_printf("\n\rStarting DA2 Pmod demo test...\n\r");
    Xil_Out32(DA2acq,0);    //stop DAC acquire

    while(1)
    {
        if(dacdata==4096)
            dacdata=0;

        dacdav=Xil_In32(DA2dav);
        if(dacdav==0 && dacacq==0)    //DAC not in use?
        {
            Xil_Out32(DA2dat1, dacdata);    //output DAC data
            Xil_Out32(DA2dat2, dacdata);    //output DAC data
            Xil_Out32(DA2acq,1);    //DAC acquire
            dacacq=1;
        }
        if(dacdav==1 && dacacq==1)    //DAC data available
        {
            Xil_Out32(DA2acq,0);    //stop DAC acquire
        }
    }
}
```

```
        dacacq=0;
        dacdata++;           //increment DAC data
    }
}
```

